2012 F=ma Corrections

Hints

December 27, 2025

Problem 1: The Dripping Faucet

To solve for the height of the drop in the air (y_2), consider the total fall time T.

- Let the time interval between drops be Δt. If the first drop hits the sink at $t = T$, then $T = 2\Delta t$.
- The total height is $H = 10$ cm. Using $y = \frac{1}{2}gt^2$, express H in terms of T.
- The second drop has been falling for a time $t = \Delta t$. Express its distance fallen d_2 as a fraction of H.
- **Critical Step:** Remember the height above the sink is $H - d_2$.

Problem 10: The Incline Race

Use the principle of Conservation of Energy: $mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$.

- Substitute $\omega = v/r$ and $I = \beta mr^2$.
- Solve for v. Does the result depend on m or r?
- Compare β for a solid sphere ($\frac{2}{5}$) and a hollow sphere ($\frac{2}{3}$).

Problem 19: The Basement Pump

The power P is the rate of change of total energy (Kinetic + Potential).

$$P = \frac{dm}{dt}\left(gh + \frac{1}{2}v^2\right) \tag{1}$$

- Recall that $\frac{dm}{dt} = \rho A v = \rho(\pi r^2)v$.
- Plug in the known values ($P = 1500, h = 2.0, v = 2.5, \rho = 1000$) and solve for r.

Problem 23: Measuring g

Evaluate each setup based on whether g can be isolated.

- In option (E), you have Work $W = mgh$. Power is $P = W/t$.
- If the string length is unknown, can you definitively determine the displacement h to solve for g?

Problem 24: Rotating Triangle

Consider the geometry of the equilateral triangle.

- Let l be the natural length and $2l$ be the extended length. The stretch Δx is l.
- The radius of rotation for a mass is $R = \frac{\text{side}}{\sqrt{3}}$.
- **Force Balance:** $F_{net} = 2F_s \cos(30^\circ) = m\omega^2 R$.
- Substitute $F_s = k(2l - l)$ and solve for k.